

Thesis: A Comparative Study of Large Language Models for Financial Sentiment Analysis and Their Predictive Potential for Short-Term Stock Price Movement

Component: Experiment 1 - Model A : FinBERT (ProsusAI/finbert)

Description: This notebook evaluates FinBERT on the Financial PhraseBank dataset under zero-shot conditions. It generates sentiment predictions and computes accuracy, precision, recall, F1-score (macro-averaged), and confusion matrix.

Select T4 GPU as runtime.

```
# Install required libraries - Experiment 1a: FinBERT

!pip install -q transformers
!pip install -q pandas scikit-learn matplotlib seaborn

print("Libraries installed successfully.")
```

Libraries installed successfully.

```
# Import required libraries - Experiment 1a: FinBERT

import pandas as pd
import numpy as np
import torch
import warnings

from transformers import pipeline
from sklearn.metrics import (
    accuracy_score, precision_score, recall_score,
    f1_score, classification_report
)

warnings.filterwarnings("ignore")

print(f"PyTorch version : {torch.__version__}")
print(f"CUDA available : {torch.cuda.is_available()}")
if torch.cuda.is_available():
    print(f"GPU detected : {torch.cuda.get_device_name(0)}")
```

PyTorch version : 2.10.0+cu128
CUDA available : True
GPU detected : Tesla T4

Add the API Key from Hugging Face using "Add New Secret" in Google Colab

```
# Connect to Hugging Face
from google.colab import userdata
from huggingface_hub import login, hf_hub_download

hf_token = userdata.get("HF_TOKEN")
login(token=hf_token)

print("Hugging Face login successful.")

# Load Financial PhraseBank - full dataset
file_path = hf_hub_download(
    repo_id="takala/financial_phrasebank",
    filename="sentences_50agree/train-00000-of-00001.parquet",
    repo_type="dataset",
    revision="0dd3028d70cbd18ded8887e65e83343b03a50482",
    token=hf_token
)

df_fpb = pd.read_parquet(file_path)

label_map = {0: "negative", 1: "neutral", 2: "positive"}
df_fpb["sentiment"] = df_fpb["label"].map(label_map)

true_labels = df_fpb["sentiment"].tolist()
texts = df_fpb["sentence"].tolist()

print(f"\nFinancial PhraseBank loaded.")
print(f"Total sentences : {len(df_fpb)}")
print(f"Label distribution:")
print(df_fpb["sentiment"].value_counts())
```

Hugging Face login successful.

sentences_50agree/train-00000-of-00001.p(...): 100%

390k/390k [00:01<00:00, 1.95MB/s]

Financial PhraseBank loaded.

Total sentences : 4846

Label distribution:

sentiment

neutral 2879

positive 1363

negative 604

Name: count, dtype: int64

```
# Load FinBERT
```

```
# Model: ProsusAI/finbert
```

```
# Domain-specific financial sentiment model
```

```
print("Loading FinBERT...")
```

```
finbert = pipeline(  
    "text-classification",  
    model="ProsusAI/finbert",  
    tokenizer="ProsusAI/finbert",  
    device=0 if torch.cuda.is_available() else -1  
)
```

```
print("FinBERT loaded successfully.")
```

Loading FinBERT...

config.json: 100%

758/758 [00:00<00:00, 43.7kB/s]

pytorch_model.bin: 100%

438M/438M [00:02<00:00, 244MB/s]

Loading weights: 100%

201/201 [00:00<00:00, 677.98it/s, Materializing param=classifier.weight]

BertForSequenceClassification LOAD REPORT from: ProsusAI/finbert

Key	Status	
bert.embeddings.position_ids	UNEXPECTED	

Notes:

- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch

model.safetensors: 100%

438M/438M [00:02<00:00, 240MB/s]

tokenizer_config.json: 100%

252/252 [00:00<00:00, 11.4kB/s]

vocab.txt: 232k/? [00:00<00:00, 4.32MB/s]

special_tokens_map.json: 100%

112/112 [00:00<00:00, 2.38kB/s]

FinBERT loaded successfully.

```
# Run FinBERT on full Financial PhraseBank (4,846 sentences)
```

```
print("Running FinBERT on 4,846 sentences...")
```

```
finbert_results = finbert(texts, batch_size=32)
```

```
finbert_preds = [r["label"].lower() for r in finbert_results]
```

```
print(f"Done. Total predictions: {len(finbert_preds)}")
```

```
print(f"\nPrediction distribution:")
```

```
print(pd.Series(finbert_preds).value_counts())
```

Running FinBERT on 4,846 sentences...

Done. Total predictions: 4846

Prediction distribution:

neutral 2567

positive 1549

negative 730

Name: count, dtype: int64

```
# Evaluate FinBERT performance
```

```
labels_order = ["positive", "negative", "neutral"]
```

```
acc = accuracy_score(true_labels, finbert_preds)
```

```
prec = precision_score(true_labels, finbert_preds, average="macro", labels=labels_order)
```

```
rec = recall_score(true_labels, finbert_preds, average="macro", labels=labels_order)
```

```
f1 = f1_score(true_labels, finbert_preds, average="macro", labels=labels_order)
```

```
print("=" * 45)
```

```
print("FinBERT - Experiment 1 Results")
```

```
print("=" * 45)
```

```
print(f" Accuracy : {acc:.4f} ({acc*100:.2f}%)")
```

```

print(f" Precision : {prec:.4f}")
print(f" Recall   : {rec:.4f}")
print(f" F1-Score  : {f1:.4f}")
print("=" * 45)

print("\nDetailed Classification Report:")
print(classification_report(true_labels, finbert_preds, labels=labels_order))

```

=====

FinBERT – Experiment 1 Results

=====

```

Accuracy : 0.8896 (88.96%)
Precision : 0.8584
Recall    : 0.9163
F1-Score  : 0.8825

```

=====

Detailed Classification Report:

	precision	recall	f1-score	support
positive	0.81	0.92	0.86	1363
negative	0.80	0.97	0.88	604
neutral	0.96	0.86	0.91	2879
accuracy			0.89	4846
macro avg	0.86	0.92	0.88	4846
weighted avg	0.90	0.89	0.89	4846

```
# Save FinBERT results and predictions
```

```
all_results = []
```

```

finbert_scores = {
    "model": "FinBERT",
    "accuracy": round(acc, 4),
    "precision": round(prec, 4),
    "recall": round(rec, 4),
    "f1_score": round(f1, 4)
}

```

```
all_results.append(finbert_scores)
```

```

# Save predictions to dataframe
df_fpb["finbert_pred"] = finbert_preds

```

```

print("FinBERT results saved.")
print(pd.DataFrame(all_results))

```

```
FinBERT results saved.
```

	model	accuracy	precision	recall	f1_score
0	FinBERT	0.8896	0.8584	0.9163	0.8825

```
# Save FinBERT predictions to Google Drive
```

```

from google.colab import drive
drive.mount("/content/drive")

```

```

import os
os.makedirs("/content/drive/MyDrive/Thesis_Data", exist_ok=True)

```

```
df_fpb.to_csv("/content/drive/MyDrive/Thesis_Data/exp1_finbert_preds.csv", index=False)
```

```
print("FinBERT predictions saved to Google Drive.")
```

```
Mounted at /content/drive
```

```
FinBERT predictions saved to Google Drive.
```

```
# Confusion Matrix – FinBERT Experiment 1
```

```

import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.metrics import confusion_matrix

```

```
labels_order = ["positive", "negative", "neutral"]
```

```
cm = confusion_matrix(true_labels, finbert_preds, labels=labels_order)
```

```

plt.figure(figsize=(7, 5))
sns.heatmap(cm, annot=True, fmt="d", cmap="Blues",
            xticklabels=labels_order,
            yticklabels=labels_order)
plt.xlabel("Predicted Label", fontsize=12)

```

```
plt.ylabel("True Label", fontsize=12)
plt.title("FinBERT — Confusion Matrix (Experiment 1)", fontsize=12)
plt.tight_layout()
plt.savefig("/content/drive/MyDrive/Thesis_Data/fig_cm_finbert.png",
            dpi=300, bbox_inches="tight")
plt.show()
print("Confusion matrix saved to Google Drive.")
```

